

FOSSTON, MN, USA

WMO: 727505

Lat: 47.593N

Lon: 95.775W

Elev: 389

StdP: 96.74

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04929

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.0	(c) -28.0	(d) -36.2	(e) 0.1	(f) -30.9	(g) -32.7	(h) 0.2	(i) -28.1	(j) 12.2	(k) -8.1	(l) 11.1	(m) -9.4	(n) 1.8	(o) 180	(p) 0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	30.9	21.7	28.7	20.2	27.4	19.5	23.3	28.1	22.0	26.9	20.8	25.7	5.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.0	17.5	27.0	20.2	15.6	25.4	18.9	14.4	24.2	71.5	28.2	66.4	26.8	62.1	26.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.0	(b) 9.4	(c) 8.4	(e) -36.1	(f) 33.0	(g) 4.6	(h) 2.1	(i) -39.4	(j) 34.5	(k) -42.1	(l) 35.7	(m) -44.7	(n) 36.9	(o) -48.1	(p) 38.5		
(a) 11.0	(b) 9.4	(c) 8.4	(e) -36.0	(f) 25.6	(g) 4.2	(h) 1.7	(i) -39.1	(j) 26.8	(k) -41.6	(l) 27.8	(m) -44.0	(n) 28.7	(o) -47.0	(p) 29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-13.9	-11.8	-3.9	4.8	12.1	17.6	19.9	18.8	14.5	6.6	-2.5	-10.5
	DBStd	13.45	8.34	7.92	7.79	6.04	5.01	3.85	3.20	3.35	5.11	5.42	6.99	7.64
	HDD10.0	3153	739	611	432	173	35	1	0	0	15	133	378	636
	HDD18.3	5271	998	844	688	406	202	58	20	34	136	364	626	894
	CDD10.0	1111	0	0	2	18	99	229	306	274	151	29	2	0
	CDD18.3	186	0	0	0	1	8	36	68	50	22	2	0	0
	CDH23.3	1794	0	0	0	7	105	337	645	477	203	19	0	0
	CDH26.7	451	0	0	0	1	27	75	170	120	53	4	0	0
Wind	WSAvg	3.5	3.8	3.6	3.7	3.9	3.9	3.2	2.7	2.8	3.4	3.9	3.9	3.8
Precipitation	PrecAvg	620	15	11	22	42	71	110	91	86	72	57	20	17
	PrecMax	843	62	32	78	143	197	294	240	247	255	258	88	47
	PrecMin	320	0	0	0	0	10	19	14	12	11	3	0	0
	PrecStd	120	14	8	16	32	39	57	54	46	45	53	18	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	7.2	17.6	23.7	29.8	31.3	32.7	32.5	31.3	26.3	16.4	7.1
		MCWB	1.7	4.6	11.8	12.6	17.6	21.8	23.0	22.4	20.8	16.2	10.2	4.6
	2%	DB	2.1	3.7	12.4	20.3	26.3	28.7	30.2	29.1	27.8	21.3	12.4	2.9
		MCWB	0.5	1.8	8.8	11.0	15.8	20.0	22.4	20.3	18.7	14.0	8.1	1.2
	5%	DB	0.0	2.1	8.6	17.5	23.0	27.2	28.0	27.6	26.0	17.7	9.0	1.3
		MCWB	-1.1	0.6	5.7	9.7	14.0	18.8	21.1	19.6	17.9	11.9	5.7	0.1
	10%	DB	-2.3	-0.2	6.1	14.8	21.2	25.2	27.2	26.2	22.9	14.9	6.3	-0.9
		MCWB	-3.3	-1.3	3.1	8.2	13.2	17.5	20.6	18.9	16.4	10.0	3.5	-1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	5.0	12.9	14.6	20.0	24.0	25.2	24.1	22.3	17.6	11.8	4.8
		MCDB	4.3	6.8	15.7	21.1	25.7	28.7	30.3	29.4	28.2	22.4	14.9	6.5
	2%	WB	0.4	2.1	8.8	12.0	17.7	22.0	23.8	22.4	20.4	14.9	8.4	1.8
		MCDB	1.5	3.8	12.5	18.0	23.3	26.6	28.3	27.7	26.0	20.3	11.3	3.2
	5%	WB	-1.2	0.4	5.6	10.4	15.9	20.2	22.6	21.1	19.1	12.4	6.1	0.1
		MCDB	-0.3	1.6	8.3	16.4	21.3	25.0	27.1	26.1	23.8	16.8	9.0	1.2
	10%	WB	-3.2	-1.3	3.4	8.7	14.4	18.9	21.3	19.9	17.7	10.6	3.8	-1.3
		MCDB	-2.4	-0.4	5.6	14.6	19.3	23.5	26.0	24.6	21.9	14.4	6.2	-0.7
Mean Daily Temperature Range	5% DB	MDBR	10.2	11.0	10.6	11.2	12.1	11.0	11.7	12.4	11.9	9.7	8.6	8.8
		MCDBR	9.1	9.9	12.8	15.7	15.8	13.4	13.2	14.1	14.2	14.1	12.0	8.9
		MCWBR	8.0	8.2	9.3	8.8	8.2	7.1	7.6	7.4	7.4	8.1	8.1	7.5
	5% WB	MCDBR	9.0	8.9	12.2	14.1	12.7	11.3	12.1	12.1	12.2	12.7	11.4	8.2
		MCWBR	8.0	7.6	9.2	8.5	7.7	7.4	7.8	7.0	7.2	7.9	8.0	7.1
Clear-Sky Solar Irradiance	taub	0.279	0.319	0.295	0.350	0.367	0.377	0.380	0.381	0.351	0.305	0.278	0.267	
	taud	2.274	2.144	2.417	2.350	2.365	2.364	2.368	2.374	2.445	2.551	2.511	2.375	
	Ebn at Noon	796	826	933	902	896	886	878	861	858	857	814	776	
	Edh at Noon	75	108	96	115	118	118	116	111	94	72	60	61	
All-Sky Solar Radiation	RadAvg	1.57	2.73	3.86	4.83	5.38	5.86	6.24	5.24	3.85	2.36	1.41	1.10	
	RadStd	0.16	0.26	0.40	0.44	0.33	0.42	0.29	0.39	0.28	0.36	0.15	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-2.43	N/A	N/A	N/A	N/A	N/A	N/A	-119	-78
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page